19 次单位根的根式解

参数与标准素分解

$p = 19$，$h = (p-1)/2 = 9$，模 19 的最小原根 $g = 2$。
h 的标准素分解：$h = 3 \times 3$，子群链共 2 层。
$m_0 = 1$，$m_1 = 3$，$m_2 = 9$。
各层子群 $H_j = \{\pm g^{m_j t}\}$，其中 H_0 是全部 18 个非零剩余：

$$H_1 = \{1, 7, 8, 11, 12, 18\}$$

$$H_2 = \{1, 18\}$$

各层周期 $\eta_a^{(j)} = \sum_{b \in H_j} \zeta^{ab}$（$a$ 取陪集代表）。

各层周期与分解表

第 0 层：1 个周期

H_0 是全部 18 个非零剩余，故 $\eta_1^{(0)} = -1$。

第 1 层：3 个周期

$$\eta_1^{(1)} = \zeta^1 + \zeta^7 + \zeta^8 + \zeta^{11} + \zeta^{12} + \zeta^{18}$$

$$\eta_2^{(1)} = \zeta^2 + \zeta^3 + \zeta^5 + \zeta^{14} + \zeta^{16} + \zeta^{17}$$

$$\eta_4^{(1)} = \zeta^4 + \zeta^6 + \zeta^9 + \zeta^{10} + \zeta^{13} + \zeta^{15}$$

第 2 层：9 个周期

$$\eta_1^{(2)} = \zeta^1 + \zeta^{18}$$

$$\eta_2^{(2)} = \zeta^2 + \zeta^{17}$$

$$\eta_3^{(2)} = \zeta^3 + \zeta^{16}$$

$$\eta_4^{(2)} = \zeta^4 + \zeta^{15}$$

$$\eta_5^{(2)} = \zeta^5 + \zeta^{14}$$

$$\eta_6^{(2)} = \zeta^6 + \zeta^{13}$$

$$\eta_7^{(2)} = \zeta^7 + \zeta^{12}$$

$$\eta_8^{(2)} = \zeta^8 + \zeta^{11}$$

$$\eta_9^{(2)} = \zeta^9 + \zeta^{10}$$

分解表：每个母周期拆成 ℓ_j 个子周期

$$\eta_1^{(0)} = \eta_1^{(1)} + \eta_2^{(1)} + \eta_4^{(1)}$$

（第 1 层，$\ell = 3$。）

$$\eta_1^{(1)} = \eta_1^{(2)} + \eta_8^{(2)} + \eta_7^{(2)}$$

$$\eta_2^{(1)} = \eta_2^{(2)} + \eta_3^{(2)} + \eta_5^{(2)}$$

$$\eta_4^{(1)} = \eta_4^{(2)} + \eta_6^{(2)} + \eta_9^{(2)}$$

（第 2 层，$\ell = 3$。）

辅助单位根

$\ell = 2$ 时 $\omega_2^1 = -1$。
$\ell \geq 3$ 时按主支：

$$\omega_\ell^k = \frac{\Lambda_{\bar{k}}^{(\ell)}}{2} + \frac{\varepsilon_k}{2}\sqrt{2 - \Lambda_{\overline{2k}}^{(\ell)}}\, i$$

其中 $\varepsilon_k = \operatorname{sgn}\sin(2\pi k/\ell)$，$\bar{k} = \min(k, \ell - k)$。

辅助问题 $p = 3$（供 ω_3^k 使用）

$h = 1$，$g = 2$，层阶 1。

$$\Lambda_1^{(3)} = -1$$

$\omega_3^k = \frac{1}{2}\Lambda_{\bar{k}}^{(3)} + \frac{\varepsilon_k}{2}\sqrt{2 - \Lambda_{\overline{2k}}^{(3)}}\, i$。

逐层开方

每一步把母周期 $\eta_a^{(j-1)}$ 分解成 ℓ 个子周期 $\beta_t^{(a)}$。
预解式取离散 Fourier 变换：

$$\theta_{a,k}^{(j)} = \sum_{t=0}^{\ell-1} \omega_\ell^{-kt}\beta_t^{(a)}$$

$$\left(\theta_{a,k}^{(j)}\right)^{\ell} = u_{a,k}^{(j)} = \tau^{k}\left(u_{a,1}^{(j)}\right)$$

$$\eta_{b(t)}^{(j)} = \frac{1}{\ell}\left(\eta_{a}^{(j-1)} + \sum_{k=1}^{\ell-1} \omega_{\ell}^{kt}\theta_{a,k}^{(j)}\right)$$

（τ 把 ω_ℓ 送到 ω_ℓ^k；$b(t)$ 为第 t 个子周期。）

开方一律取正实主支；预解式的分支由因子 ω_ℓ^s 记录（见报告里的分支表）。

第 1 步（$\ell = 3$，**1 个母周期**）

$$u_{1,1}^{(1)} = \frac{133 + 57i\sqrt{3}}{2}$$

$$\theta_{1,1}^{(1)} = \frac{\left(-1 - i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}}}{4}$$

第 2 步（$\ell = 3$，**3 个母周期**）

$$u_{1,1}^{(2)} = \frac{152 + \left(23 + 3i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(14 + 6i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}{12}$$

$$\theta_{1,1}^{(2)} = \frac{\sqrt[3]{2736 + \left(81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} + 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(414 - 27i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(252 + 27i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}}{6}$$

$$u_{2,1}^{(2)} = \frac{76 + \left(-8 + 5i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(1 - 5i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}{6}$$

$$\theta_{2,1}^{(2)} = \frac{\sqrt[3]{2736 + \left(-288 + 180i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(36 - 180i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}}{6}$$

$$u_{4,1}^{(2)} = \frac{152 + \left(-7 - 13i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-16 + 4i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}{12}$$

$$\theta_{4,1}^{(2)} = \frac{\sqrt[3]{2736 + \left(-81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} - 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(-126 - 153i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-288 + 153i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}}{6}$$

实部数值

$$\Lambda_1 = 2\cos\frac{2\pi\cdot 1}{19} = \frac{-4 + \left(-1 - i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-1 + i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}} + 2\sqrt{2736 + \left(81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} + 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(414 - 27i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(252 + 27i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}} + 2\sqrt{2736 + \left(-81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} - 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(252 - 27i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(414 + 27i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}}{36}$$

$$\Lambda_2 = 2\cos\frac{2\pi\cdot 2}{19} = \frac{-2 + \sqrt[3]{532 + 228i\sqrt{3}} + \sqrt[3]{532 - 228i\sqrt{3}} + \sqrt[3]{2736 + \left(-288 + 180i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(36 - 180i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}} + \sqrt[3]{2736 + \left(36 + 180i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-288 - 180i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}}{18}$$

$$\Lambda_3 = 2\cos\frac{2\pi\cdot 3}{19} = \frac{-4 + 2\sqrt[3]{532 + 228i\sqrt{3}} + 2\sqrt[3]{532 - 228i\sqrt{3}} + \left(-1 + i\sqrt{3}\right)\sqrt[3]{2736 + \left(-288 + 180i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(36 - 180i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}} + \left(-1 - i\sqrt{3}\right)\sqrt[3]{2736 + \left(36 + 180i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-288 - 180i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}}{36}$$

$$\Lambda_4 = 2\cos\frac{2\pi\cdot 4}{19} = \frac{-4 + \left(-1 + i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-1 - i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}} + 2\sqrt{2736 + \left(-81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} - 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(-126 - 153i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-288 + 153i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}} + 2\sqrt{2736 + \left(81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} + 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(-288 - 153i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-126 + 153i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}}{36}$$

$$\Lambda_5 = 2\cos\frac{2\pi\cdot 5}{19} = \frac{-4 + 2\sqrt[3]{532 + 228i\sqrt{3}} + 2\sqrt[3]{532 - 228i\sqrt{3}} + \left(-1 - i\sqrt{3}\right)\sqrt[3]{2736 + \left(-288 + 180i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(36 - 180i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}} + \left(-1 + i\sqrt{3}\right)\sqrt[3]{2736 + \left(36 + 180i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-288 - 180i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}}{36}$$

$$\Lambda_6 = 2\cos\frac{2\pi\cdot 6}{19} = \frac{-4 + \left(-1 + i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-1 - i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}} + \left(-1 + i\sqrt{3}\right)\sqrt{2736 + \left(-81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} - 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(-126 - 153i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-288 + 153i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}} + \left(-1 - i\sqrt{3}\right)\sqrt{2736 + \left(81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} + 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(-288 - 153i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-126 + 153i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}}{36}$$

$$\Lambda_7 = 2\cos\frac{2\pi\cdot 7}{19} = \frac{-4 + \left(-1 - i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-1 + i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}} + \left(-1 - i\sqrt{3}\right)\sqrt{2736 + \left(81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} + 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(414 - 27i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(252 + 27i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}} + \left(-1 + i\sqrt{3}\right)\sqrt{2736 + \left(-81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} - 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(252 - 27i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(414 + 27i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}}{36}$$

$$\Lambda_8 = 2\cos\frac{2\pi\cdot 8}{19} = \frac{-4 + \left(-1 - i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-1 + i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}} + \left(-1 + i\sqrt{3}\right)\sqrt{2736 + \left(81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} + 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(414 - 27i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(252 + 27i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}} + \left(-1 - i\sqrt{3}\right)\sqrt{2736 + \left(-81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} - 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(252 - 27i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(414 + 27i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}}{36}$$

$$\Lambda_9 = 2\cos\frac{2\pi\cdot 9}{19} = \frac{-4 + \left(-1 + i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-1 - i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}} + \left(-1 - i\sqrt{3}\right)\sqrt{2736 + \left(-81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} - 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(-126 - 153i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-288 + 153i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}} + \left(-1 + i\sqrt{3}\right)\sqrt{2736 + \left(81\sqrt{3}\sqrt[3]{532 + 228i\sqrt{3}} + 81\sqrt{3}\sqrt[3]{532 - 228i\sqrt{3}}\right)i + \left(-288 - 153i\sqrt{3}\right)\sqrt[3]{532 + 228i\sqrt{3}} + \left(-126 + 153i\sqrt{3}\right)\sqrt[3]{532 - 228i\sqrt{3}}}}{36}$$

全部单位根

最终形式：

$$\omega_n^t = \frac{\Lambda_{\bar{t}}}{2} + \frac{\varepsilon_t}{2}\sqrt{2-\Lambda_{\overline{2t}}}\,i, \qquad \Lambda_{\bar{t}} = 2\cos\frac{2\pi\bar{t}}{n}$$

其中 $\bar{t} = \min(t, n-t)$，$\varepsilon_t = \operatorname{sgn}\sin(2\pi t/n)$。

以下逐一写开（全部为具体根式）：

$$\omega_{19}^{1} = \frac{-4 + \left(-1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}} + 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(311 - 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(253 + 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(253 + 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(311 - 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 6i\sqrt{76 - 2\sqrt{532+228\sqrt{5}} - 2\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}{72}$$

$$\omega_{19}^{2} = \frac{-2 + \sqrt{532+228\sqrt{5}} + \sqrt{532-228\sqrt{5}} + \sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 3i\sqrt{76 + \left(1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-126 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-266 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-126 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{36}$$

$$\omega_{19}^{3} = \frac{-2 + \sqrt{532+228\sqrt{5}} + \sqrt{532-228\sqrt{5}} + \left(-1+\sqrt{5}\right)\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(-1-\sqrt{5}\right)\sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 6i\sqrt{76 + \left(1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(1-\sqrt{5}\right)\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-126 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(1+\sqrt{5}\right)\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-266 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-126 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{72}$$

$$\omega_{19}^{4} = \frac{-2 + \sqrt{532+228\sqrt{5}} + \sqrt{532-228\sqrt{5}} + \left(-1+\sqrt{5}\right)\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(-1-\sqrt{5}\right)\sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 6i\sqrt{76 + \left(1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(1-\sqrt{5}\right)\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-126 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(1+\sqrt{5}\right)\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-266 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-126 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{72}$$

$$\omega_{19}^{5} = \frac{-4 + \left(1+\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1-\sqrt{5}\right)\sqrt{532-228\sqrt{5}} + 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(311 - 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(253 + 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(253 + 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(311 - 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 6i\sqrt{76 - 2\sqrt{532+228\sqrt{5}} - 2\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{72}$$

$$\omega_{19}^{6} = \frac{-2 + \sqrt{532+228\sqrt{5}} + \sqrt{532-228\sqrt{5}} + \sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 3i\sqrt{76 + \left(1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-126 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-266 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-126 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{36}$$

$$\omega_{19}^{7} = \frac{-4 + \left(-1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}} + 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(311 - 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(253 + 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(253 + 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(311 - 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 6i\sqrt{76 - 2\sqrt{532+228\sqrt{5}} - 2\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{72}$$

$$\omega_{19}^{8} = \frac{-2 + \sqrt{532+228\sqrt{5}} + \sqrt{532-228\sqrt{5}} + \left(-1+\sqrt{5}\right)\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(-1-\sqrt{5}\right)\sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 6i\sqrt{76 + \left(1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(1-\sqrt{5}\right)\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-126 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(1+\sqrt{5}\right)\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-266 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-126 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{72}$$

$$\omega_{19}^{9} = \frac{-4 + \left(1+\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1-\sqrt{5}\right)\sqrt{532-228\sqrt{5}} + 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(311 - 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(253 + 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(253 + 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(311 - 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 6i\sqrt{76 - 2\sqrt{532+228\sqrt{5}} - 2\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{72}$$

$$\omega_{19}^{10} = \frac{-2 + \sqrt{532+228\sqrt{5}} + \sqrt{532-228\sqrt{5}} + \sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 3i\sqrt{76 + \left(1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-126 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-266 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-126 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{36}$$

$$\omega_{19}^{11} = \frac{-2 + \sqrt{532+228\sqrt{5}} + \sqrt{532-228\sqrt{5}} + \left(-1+\sqrt{5}\right)\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(-1-\sqrt{5}\right)\sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 6i\sqrt{76 + \left(1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(1-\sqrt{5}\right)\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-126 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(1+\sqrt{5}\right)\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-266 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-126 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{72}$$

$$\omega_{19}^{12} = \frac{-4 + \left(1+\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1-\sqrt{5}\right)\sqrt{532-228\sqrt{5}} + 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(311 - 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(253 + 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(253 + 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(311 - 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 6i\sqrt{76 - 2\sqrt{532+228\sqrt{5}} - 2\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{72}$$

$$\omega_{19}^{13} = \frac{-2 + \sqrt{532+228\sqrt{5}} + \sqrt{532-228\sqrt{5}} + \sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 3i\sqrt{76 + \left(1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-126 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-266 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-126 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{36}$$

$$\omega_{19}^{14} = \frac{-4 + \left(-1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}} + 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(311 - 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(253 + 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(253 + 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(311 - 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 6i\sqrt{76 - 2\sqrt{532+228\sqrt{5}} - 2\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{72}$$

$$\omega_{19}^{15} = \frac{-4 + \left(1+\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1-\sqrt{5}\right)\sqrt{532-228\sqrt{5}} + 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(311 - 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(253 + 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(253 + 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(311 - 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 6i\sqrt{76 - 2\sqrt{532+228\sqrt{5}} - 2\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{72}$$

$$\omega_{19}^{16} = \frac{-2 + \sqrt{532+228\sqrt{5}} + \sqrt{532-228\sqrt{5}} + \sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 3i\sqrt{76 + \left(1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-126 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-266 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-126 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{36}$$

$$\omega_{19}^{17} = \frac{-4 + \left(-1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}} + 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(311 - 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(253 + 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(253 + 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(311 - 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 6i\sqrt{76 - 2\sqrt{532+228\sqrt{5}} - 2\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{72}$$

$$\omega_{19}^{18} = \frac{-2 + \sqrt{532+228\sqrt{5}} + \sqrt{532-228\sqrt{5}} + \left(-1+\sqrt{5}\right)\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(-1-\sqrt{5}\right)\sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 6i\sqrt{76 + \left(1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(1-\sqrt{5}\right)\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-126 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + \left(1+\sqrt{5}\right)\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(-266 - 153\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-126 + 153\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{72}$$

$$\omega_{19}^{19} = \frac{-4 + \left(-1-\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-1+\sqrt{5}\right)\sqrt{532-228\sqrt{5}} + 2\sqrt{2736 + \left(61\sqrt{5}\sqrt{532+228\sqrt{5}} + 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(311 - 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(253 + 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(-61\sqrt{5}\sqrt{532+228\sqrt{5}} - 81\sqrt{5}\sqrt{532-228\sqrt{5}}\right)i + \left(253 + 271\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(311 - 271\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} - 6i\sqrt{76 - 2\sqrt{532+228\sqrt{5}} - 2\sqrt{532-228\sqrt{5}}} - 2\sqrt{2736 + \left(-266 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(36 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}} + 2\sqrt{2736 + \left(36 + 180\sqrt{5}\right)\sqrt{532+228\sqrt{5}} + \left(-266 - 180\sqrt{5}\right)\sqrt{532-228\sqrt{5}}}}{72}$$